

scRNA-seq Marker Genes Cheat Sheet

A context-aware quick reference for initial cell-type annotation. Compiled by Nasir Mahmood Abbasi, PhD.

Use this guide correctly

Marker genes are supporting evidence, not labels by themselves. Confirm annotations using multiple positive and negative markers, cluster-level differential expression, tissue and species context, cell state, sample metadata, and-where available-reference mapping or orthogonal assays. Do not use this sheet for clinical diagnosis or clinical decision-making.

Common immune populations

These marker sets are representative examples for common human immune contexts. Expression can vary by tissue, activation, disease, dissociation protocol, and sequencing depth.

Cell population	Representative markers
T cells (general)	CD3D, CD3E, CD3G, TRBC1/TRBC2
CD4+ T cells	IL7R, LTB, CCR7, MALAT1; confirm with CD4 where detected
CD8+ / cytotoxic T cells	CD8A, CD8B, NKG7, CCL5, GZMK/GZMB depending on state
NK cells	NKG7, GNLY, KLRD1, PRF1, FCGR3A
B cells	MS4A1, CD79A, CD79B, CD74, HLA-DRA
Plasma cells	JCHAIN, MZB1, XBP1, SDC1, immunoglobulin genes
CD14+ monocytes	LYZ, S100A8, S100A9, CTSD, FCN1
FCGR3A+ monocytes	FCGR3A, LST1, MS4A7, IFITM3, LILRB1
Macrophages	C1QC, APOC1, LGMN, CD68, CD163; context dependent
Conventional dendritic cells	CLEC10A, FCER1A, CD1C; cDC1 may express CLEC9A
Plasmacytoid dendritic cells	GZMB, IRF7, IL3RA, TCF4, SERPINF1; use a panel and context

Stromal, vascular, and epithelial context

Use broad lineage markers as a starting point, then examine tissue-specific programs and technical artifacts.

Cell population	Representative markers
Endothelial cells	PECAM1, VWF, KDR, EMCN; assess vessel subtype markers in context
Fibroblasts	COL1A1, COL1A2, DCN, LUM, COL3A1; state programs can vary strongly
Epithelial cells	EPCAM, KRT8, KRT18, KRT19; use tissue-specific epithelial markers
Cycling cells	MKI67, TOP2A, HMGB2; treat as a state overlay, not a lineage
Stressed / dissociation response	FOS, JUN, HSP genes, DDIT4; assess technical context before labeling
Doublet warning	Coexpression of incompatible lineage programs; inspect UMI counts, doublet scores, and cluster context

Annotation checklist

1. Inspect cluster markers and the full expression pattern, not one gene. 2. Check tissue, species, condition, and expected lineage composition. 3. Compare multiple reference sources or classifiers. 4. Record a confidence level and competing hypotheses. 5. Validate important labels with additional markers, spatial context, protein data, or orthogonal experiments when possible.

Further resources

Single Cell Best Practices: sc-best-practices.org. CellMarker: bio-bigdata.hrbmu.edu.cn/CellMarker. PanglaoDB: panglaoDB.se. Scanpy and Seurat documentation provide workflow-specific annotation examples. Always record the database/version and access date used for a research analysis.

More tutorials and practical workflows: theomicshub.com